

PKCERT AI & Software Development Internship

Task 16 – End-to-End Machine Learning Pipeline Project

Objective:

Develop a complete end-to-end machine learning solution by applying concepts learned throughout the internship. Build a complete ML pipeline from data preparation to model development, evaluation, and project documentation.

Total Marks: 100

Part A – Dataset Selection & Data Preparation (25 Marks)

1. Select a publicly available dataset.
2. Describe the dataset, features, and target variable.
3. Perform data cleaning, preprocessing, feature engineering, and exploratory data analysis.

Part B – Model Development (35 Marks)

1. Develop an end-to-end machine learning pipeline.
2. Train and evaluate one or more suitable machine learning models.
3. Apply appropriate preprocessing and feature transformation techniques.

Part C – Model Evaluation (20 Marks)

1. Evaluate the final model using suitable performance metrics.
2. Interpret the results and discuss strengths and limitations.

Part D – Project Documentation (20 Marks)

1. Organize the project using clean, readable, and well-documented code.
2. Prepare a comprehensive README describing the dataset, methodology, implementation, results, and conclusion.

Assessment Area	Marks
Dataset Selection & Preparation	25
Model Development	35
Model Evaluation	20
Project Documentation	20
Total	100